CLASS 9 · CBSE · GANITA MANJARI PART 2 · FORMULA BANK

Chapter 9: Propositions and their Converses

Every formula and result of the chapter on one sheet

1	Proposition	A statement that is either true or false
2	If X, then Y	Also written "X implies Y" or "Y when X"
3	Converse of "If X, then Y"	"If Y, then X"
4	Counterexample	A case where the condition holds but the conclusion fails
5	To disprove a proposition	Give one counterexample
6	To prove a proposition	Give a general argument, not examples
7	X and Y imply each other	Both "If X, then Y" and "If Y, then X" are true
8	Baudhāyana–Pythagoras theorem	If a triangle is right-angled, then $a^2 + b^2 = c^2$
9	Converse of the theorem	If $a^2 + b^2 = c^2$, then the triangle is right-angled
10	Perfect squares	n is a perfect square exactly when it has an odd number of factors
11	Isosceles triangle	Two sides are equal exactly when the angles opposite them are equal